

1. What is Exploratory Data Analysis (EDA)?

- EDA is an approach to analysing datasets to summarize their main characteristics, often using graphical techniques and summary statistics.

2. Why is EDA important?

- EDA helps in understanding the data, finding patterns, detecting anomalies, and preparing for further analysis or modelling.

3. How do you handle missing values during EDA?

- Missing values can be handled by imputation (replacing missing values with estimated ones), deletion (removing rows or columns with missing values), or using advanced techniques like predictive modelling to fill missing values.

4. What techniques do you use for data visualization in EDA?

- Common techniques include histograms, box plots, scatter plots, bar charts, heatmaps, and pair plots.

5. What is the purpose of a histogram in EDA?

- Histograms display the distribution of a continuous variable by dividing the data into intervals (bins) and showing the frequency of observations in each bin.

6. Explain the concept of outliers in EDA. How do you identify and handle them?

- Outliers are data points that significantly deviate from the rest of the dataset. They can be identified using statistical methods such as the IQR method or visualization techniques such as box plots. Depending on the scenario, outliers can be treated by removing them, transforming the data, or using robust statistical techniques.

7. What is correlation analysis? How is it useful in EDA?

- Correlation analysis measures the strength and direction of the linear relationship between two variables. It helps in identifying associations between variables and understanding their dependencies.

8. How do you deal with categorical variables in EDA?

- Categorical variables can be analyzed using frequency tables, bar plots, and box plots. Techniques such as one-hot encoding or label encoding may be applied for further analysis specially in case of machine learning/predictive analytics applications.

9. What is the purpose of a box plot in EDA?

- Box plots display the distribution of a continuous variable and highlight key statistics such as the median, quartiles, and outliers.

10. Explain the concept of skewness and kurtosis in EDA.

- Skewness measures the asymmetry of the distribution of a variable, while kurtosis measures the peakedness or flatness of the distribution.

11. How do you assess the normality of a variable's distribution during EDA?

- Normality can be assessed visually using histograms or Q-Q plots.

12. What is the purpose of a scatter plot in EDA?

- Scatter plots show the relationship between two continuous variables and help in identifying patterns such as trends, clusters, or correlations.

13. Explain the concept of data transformation in EDA.

- Data transformation involves converting variables to a different scale or form to meet the assumptions of statistical methods or improve the interpretability of the data.

14. What are some common tools and libraries used for EDA?

- Common tools include, Excel, SQL, Tableau and Python libraries like Pandas, NumPy, Matplotlib, Seaborn.

15. What is the purpose of a correlation matrix in EDA?

- A correlation matrix displays the correlation coefficients between pairs of variables, helping to identify relationships and dependencies among variables.

16. How do you detect data distribution anomalies during EDA?

- Data distribution anomalies can be detected by visual inspection of histograms, box plots, or by performing statistical tests for normality.

17. Explain the concept of data aggregation in EDA.

- Data aggregation involves summarizing data at a higher level of granularity, such as computing means, sums, or counts over subsets of the data.

18. What are some common challenges encountered during EDA?

- Common challenges include dealing with missing values, outliers, noisy data.

19. What are some techniques for outlier detection during EDA?

- Techniques include z-score method, IQR method, isolation forests, and clustering-based approaches.

20. How do you choose the appropriate visualization techniques for EDA?

- The choice of visualization techniques depends on the type of data (e.g., continuous, categorical), the relationships you want to explore, and the insights you want to convey.

21. How do you explore time-series data during EDA?

- Time-series data can be explored using line plots.

22. How do you handle data scaling during EDA?

- Data scaling techniques such as min-max scaling or standardization (z-score normalization) can be applied to ensure that all variables are on a similar scale.

23. What is the role of domain knowledge in EDA?

- Domain knowledge helps in interpreting the data, identifying relevant features, understanding the context of the analysis, and guiding the selection of appropriate techniques.

24. How do you communicate the findings from EDA to stakeholders?

- Findings from EDA can be communicated through clear and concise visualizations, summary statistics, and narrative explanations, tailored to the audience's level of understanding and objectives.

25. What is the role of hypothesis testing in EDA?

- Hypothesis testing helps in making statistical inferences about the data, such as comparing means or testing for associations between variables.